Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df001-2017-drone-sd
Operator:	Floris
Analysis Started:	2026-06-16 11:41:50
Analysis Completed:	2026-06-16 11:42:34
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2017
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\dfdof_output\CASE-df001-2017-drone-sd
Evidence Sources:	1
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df001_external.001	drone_sd	physical	7.4 GB	2026-06-16 11:42:24

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2017

Cryptographic Hashes

df001_external.001	drone_sd	physical
SHA-256:	7d1392046d20ef9052e4ac552791efae68a130d8893dfe4dbcba456280f98006	
SHA-1:	7b4451c3a65f8c0c666739031e58c90c2b81e8b2	

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p2 - controller ios]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios]: source evidence not found
5. [p3 - drone flight storage]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - controller ios]: source evidence not found
8. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	4	3	[~] Partial
fls	1	1	[OK]
icat	4	4	[OK]
mmls	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	1 / 4
Artefact Categories With Data	2 / 7
Flights With Primary Correlation	0 / 0
Tools Succeeded	3 / 4

3. Device and Account Identification

No device or account information could be extracted.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
drone_sd	0	0	0	0	0	3	1

Data Quality Notes per Category

Images (3 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 3 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

Videos (1 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 1 file(s) missing (normalisable) EXIF timestamp

6. Data Quality and Anomalies

Data Quality Observations

No data quality issues recorded.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p2 - controller ios]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios]: source evidence not found
5. [p3 - drone flight storage]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - controller ios]: source evidence not found
8. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Images

DJI_0001.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Note: no drone serial number found - Device and account information.
4. Note: no drone model found - Device and account information.
5. Note: no controller device found - Device and account information.
6. Note: no dji application version found - Device and account information.
7. Note: no account e-mail found - Device and account information.
8. Note: no product type found - Device and account information.
9. Note: no firmware version found - Device and account information.
10. Note: no last country code found - Device and account information.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

DJI_0001.JPG		Images	drone_sd	extract_physical
SHA-256:	d71919439f8598a703cd1b66a2e74db5151d04995079234cc079d7758c1d0ce9			
SHA-1:	27c2ae11b32f1f8682362614de078e0c7067f4f5			

DJI_0002.JPG		Images	drone_sd	extract_physical
SHA-256:	a79633b436b3b15cd7d558f62fb1fc87ffd8e24ebbc9920f3ef1a95555f7f7eb			
SHA-1:	5d7c7d32dd574871f91ce593f8cc93aef554b927			

DJI_0003.THM		Images	drone_sd	extract_physical
SHA-256:	344ec21e788f5699ec63d541f45bf07b5b0d05eb2edfada3da01d3495fbce2d2			
SHA-1:	2bc16fd536e03cbb601b679802d24de747f36b9a			

DJI_0003.MOV		Videos	drone_sd	extract_physical
SHA-256:	f19d2580881d7bdde3fb3026d609e856d456be91c101b494a208708562c24f7e			
SHA-1:	510256d33e0d12ba9838972971202885459f3e55			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	3	3	0
Videos	1	1	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-16 11:41:50 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df001_external.001

2026-06-16 11:42:00 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df001_external.001

2026-06-16 11:42:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df001_external.001 107525

out: DJI_0001.JPG

2026-06-16 11:42:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df001_external.001 107526

out: DJI_0002.JPG

2026-06-16 11:42:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df001_external.001 107527

out: DJI_0003.MOV

2026-06-16 11:42:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df001_external.001 868357

out: DJI_0003.THM

2026-06-16 11:42:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.JPG

out: DJI_0001_exif.json

2026-06-16 11:42:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG

out: DJI_0002_exif.json

2026-06-16 11:42:33 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM

out: DJI_0003_exif.json

2026-06-16 11:42:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV

out: DJI_0003_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.